

Kunal Prajapati

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SUMMARY

Radiation Physicist with expertise in radiation dosimetry, safety compliance, and particle accelerator-based systems, including proton therapy. Experienced in radiation field measurements, personnel dosimetry, and regulatory compliance in high-energy environments. Proven ability to implement radiation protection programs and ensure safe operation of complex radiation facilities.

EDUCATION

Homi Bhabha National Institute, BARC <i>Post MSc. Diploma in Radiological Physics</i>	Aug 2020 85.17%
Indian Institute of Technology, Delhi <i>Master of Science (MS) in Physics</i>	May 2019 CGPA 7.725
Panjab University <i>Bachelor of Science (BS)</i>	May 2016 78.55%

CERTIFICATIONS

• Radiation safety officer (RSO), level III, <i>Atomic Energy Regulatory Board, India</i>	Dec 2022
• CS50's Introduction to Programming with Python, <i>Harvard University</i>	Dec 2024
• Python for Data Science, <i>National Program on Technology Enhanced Learning (NPTEL)</i>	Aug 2022
• Accelerator Physics, <i>National Program on Technology Enhanced Learning (NPTEL)</i>	Oct 2022
• Medical Image Analysis, <i>National Program on Technology Enhanced Learning (NPTEL)</i>	Oct 2022
• Good clinical practice, <i>NIDA Clinical Trials Network</i>	Jan 2023

SKILLS

• Technical: Machine learning MATLAB (beginner)	Python/Scikit-learn Origin	Data Visualization SQL	HTML/CSS 3D Printing (CAD)
• Languages: English (Fluent)	Hindi (Native)	Punjabi (Fluent)	

WORK EXPERIENCE

IBA Proton Therapy, TMC Mumbai <i>Beam Physics Service Engineer & Radiation Safety Officer</i>	Navi Mumbai May 2023 – Present
- Implemented radiation safety programs and documentation to ensure compliance with AERB regulations. - Developed and managed the Radiation Protection Program for proton therapy operations - Performed radiation surveys, personnel dosimetry assessments, and dose optimization activities. - Led radiation safety audits and supported cyclotron maintenance in activated environments. - Managed radioactive waste handling, instrument calibration, and occupational dose record systems.	

Advanced Centre for Treatment, Research, and Education in Cancer (ACTREC)

Navi Mumbai

Medical Physicist

Feb 2022 – May 2023

- Conducted treatment planning and QA for advanced radiotherapy techniques like VMAT, TBI, TSET, TMLI.
- Contributed to clinical research in Medical Physics and Radiation Oncology.
- Performed HDR Brachytherapy planning and QA.

Tata Memorial Hospital, Mumbai

Mumbai

Resident Medical Physicist (internship)

Jan 2021 – Dec 2021

- Conducted treatment planning and QA for different radiotherapy techniques.
- Carried out HDR brachytherapy planning and QA.

HANDS-ON EXPERIENCE WITH RADIOTHERAPY SYSTEMS

- **Proton Therapy System**

IBA Proteus PLUS

- **LINACs**

Varian Truebeam

Varian Novalis Tx

Varian Trilogy

Accuray Tomotherapy

- **Brachytherapy Units**

Elekta microSelectron V3 afterloading system

Elekta Flexitron afterloading system

PUBLICATIONS

- **Sigmoid dose accumulation and reporting for multifractionated brachytherapy for cervical cancer**
Bindal, M. A., Mittal, P., Shinghal, A., Scaria, L., **Prajapati, K.**, et al. (2021)
- **Spatially fractionated radiotherapy using straight and bending virtual rods as an alternative for cervical cancer brachytherapy—results from preclinical in-silico dose comparison and quality assurance**
Shinde, P., A. K., Agarwal, P., Kamalnath, J., Joshi, K., **Prajapati, K.**, et al. (2026)
- **Patterns of relapse after adjuvant (chemo)radiation for cervical cancer in a phase III clinical trial (PARCER): an evaluation of updated NRG Oncology /RTOG target delineation guidelines**
Mittal, P., Chopra, S., Charnalia, M., Dora, T., Engineer, R., **Prajapati, K.**, et al. (2022)

ABSTRACTS

- **Validation of sigmoid points for sigmoid dose accumulation and reporting for multifractionated brachytherapy for cervical cancer**
Kamalnath, J., A. K., **Prajapati, K.**, et al. (2024)
- **Sigmoid Points predict Late Toxicity Assessment for Multifractionated Brachytherapy for Cervical Cancer**
Mittal P., Kamalnath, J., A. K., **Prajapati, K.**, et al. (2024)

PROFESSIONAL AFFILIATIONS

Association of Medical Physicists in India (AMPI), *Medical Physicist*

Life-time member